

BioHarvest-Sim: Analytical Assessment Report

Species: Grizzly Bear | Generated: 2026-09-07 12:02

ISO/IEC/IEEE 12207 Compliance | High-Precision Linear Algebra Core

EXECUTIVE SUMMARY:

This report details the demographic stability and sustainable yield optimization for 'Grizzly Bear' based on discrete-time Leslie matrix theory. The population exhibits a dominant eigenvalue $\lambda_1 = 0.9345$, corresponding to an intrinsic growth rate of $r = -0.0678$ per year. The net reproductive rate R_0 is 0.6627 with a mean generation time of 6.07 years. The theoretical maximum uniform sustainable harvest fraction is 0.00% per annum.

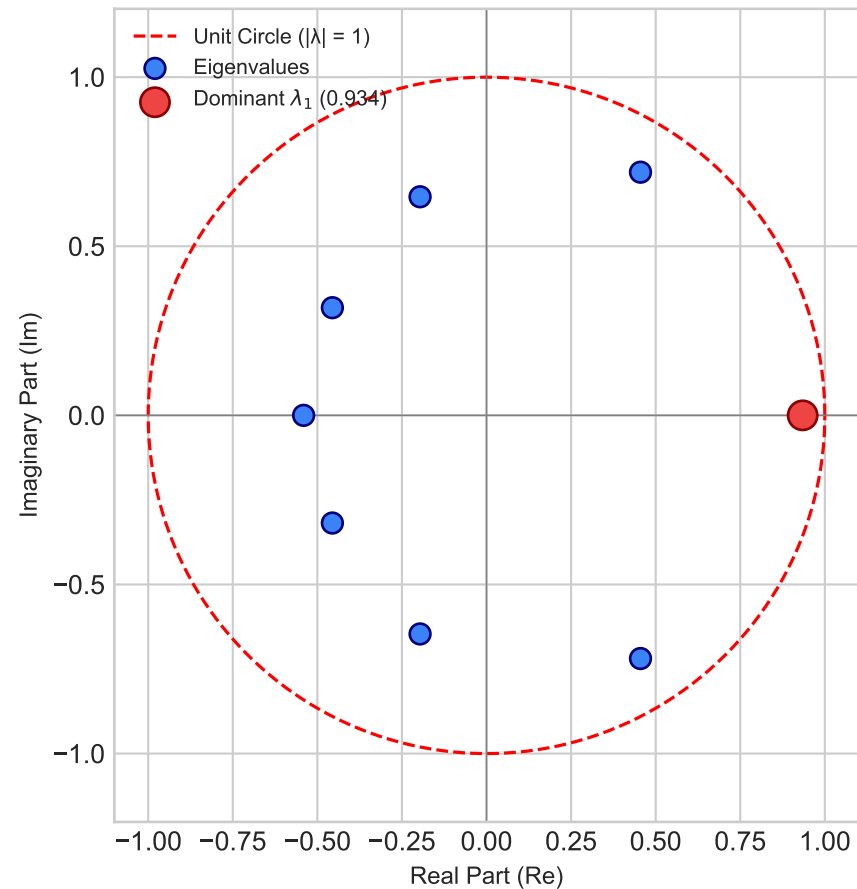
Age Classes (N)	8
Dominant Eigenvalue (λ_1)	0.93447
Intrinsic Growth Rate ($r = \ln \lambda_1$)	-0.06777 yr ⁻¹
Net Reproductive Rate (R_0)	0.66270
Mean Generation Time (Tc)	6.07 years
Damping Ratio ($\lambda_1 / \lambda_2 $)	1.0982
Perron-Frobenius Primitivity	Verified Primitive
Matrix Diagonalizable	Yes (L = P D P ⁻¹)
Uniform Sustainable Harvest (h*)	0.00% per step

Leslie Projection Matrix (L):

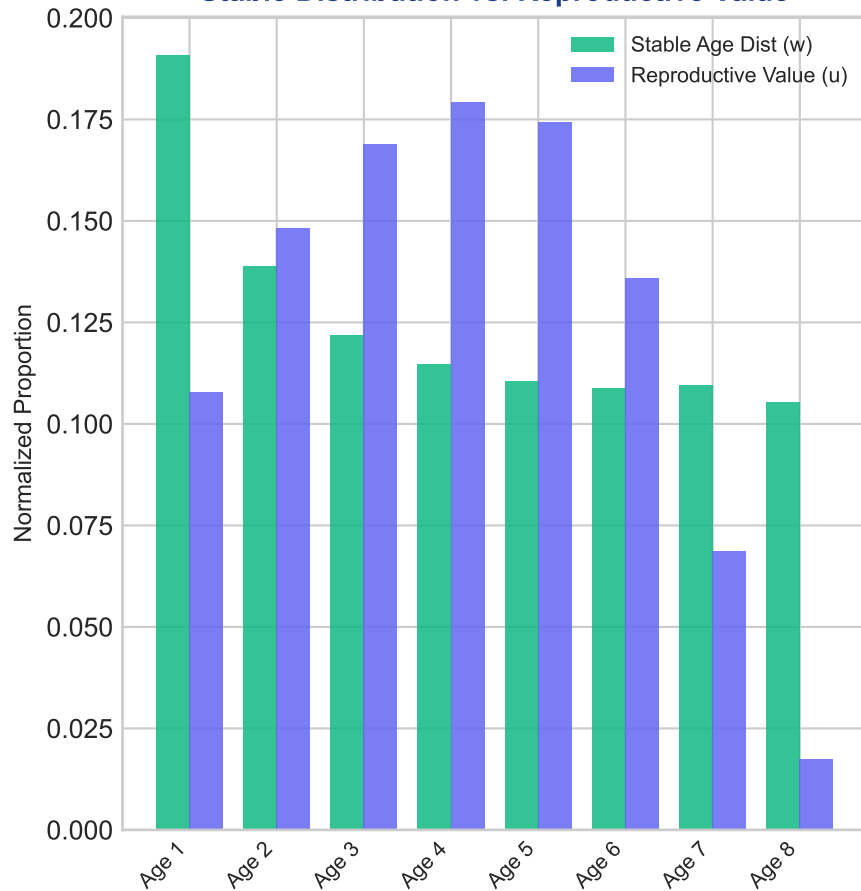
[	[	0.	0.	0.	0.1	0.35	0.58	0.45	0.15]
	[	0.68	0.	0.	0.	0.	0.	0.	]
	[	0.	0.82	0.	0.	0.	0.	0.	]
	[	0.	0.	0.88	0.	0.	0.	0.	]
	[	0.	0.	0.	0.9	0.	0.	0.	]
	[	0.	0.	0.	0.	0.92	0.	0.	]
	[	0.	0.	0.	0.	0.	0.94	0.	]
	[	0.	0.	0.	0.	0.	0.	0.9	]]

Spectral Decomposition & Asymptotic Age Structure

Eigen-Spectrum (Complex Plane)

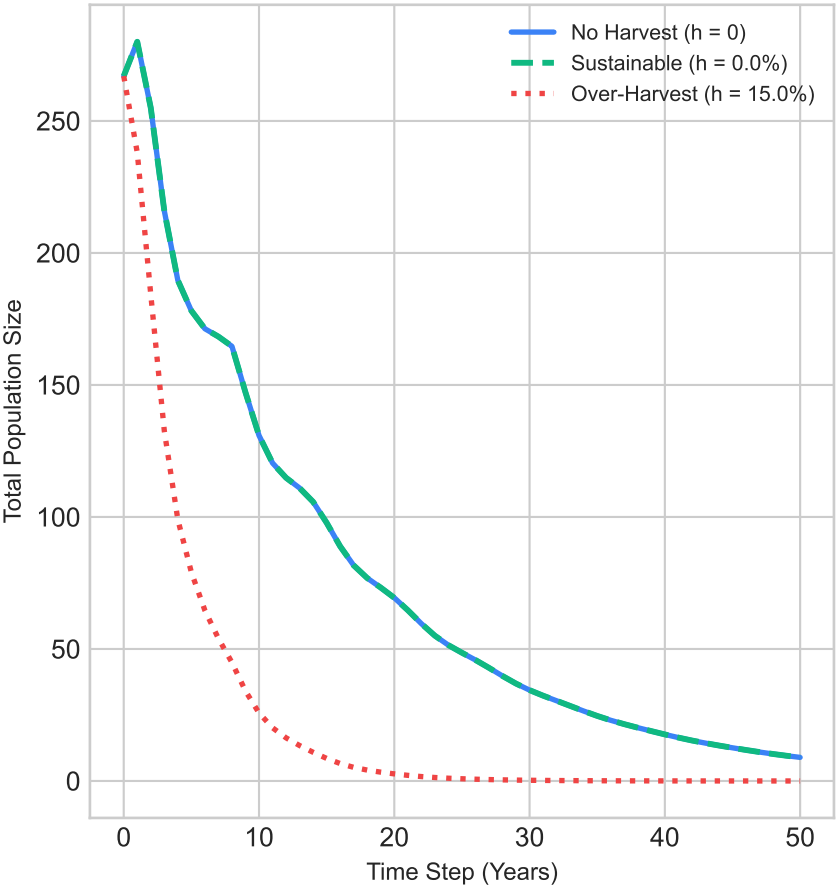


Stable Distribution vs. Reproductive Value

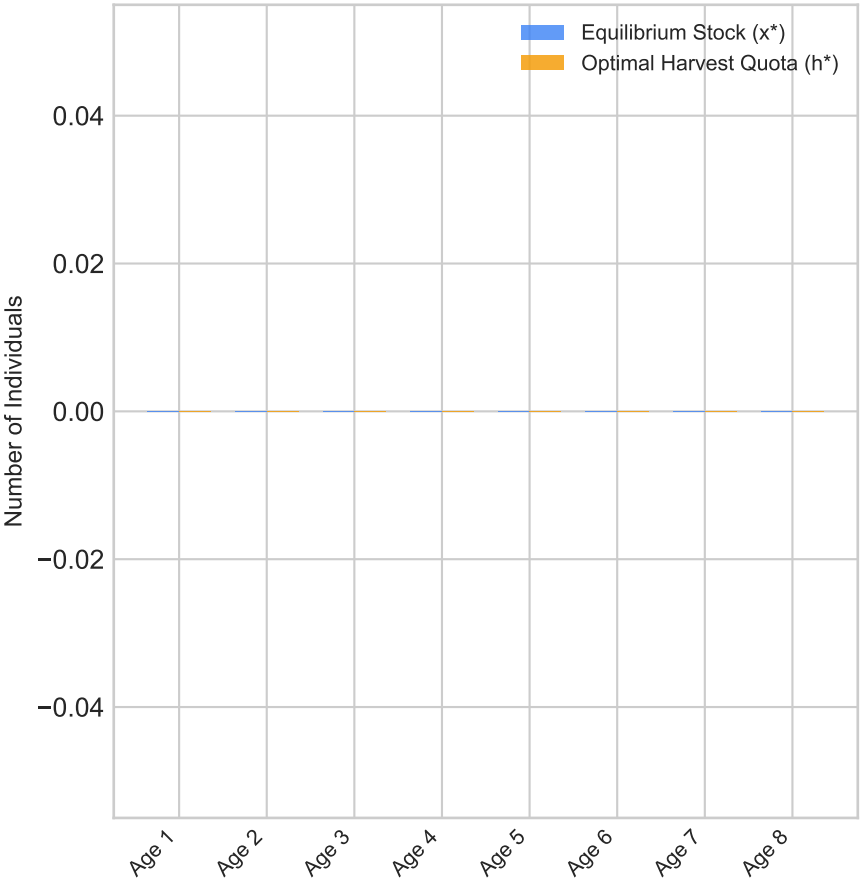


Harvest Dynamics & Optimization

Population Trajectory Comparison

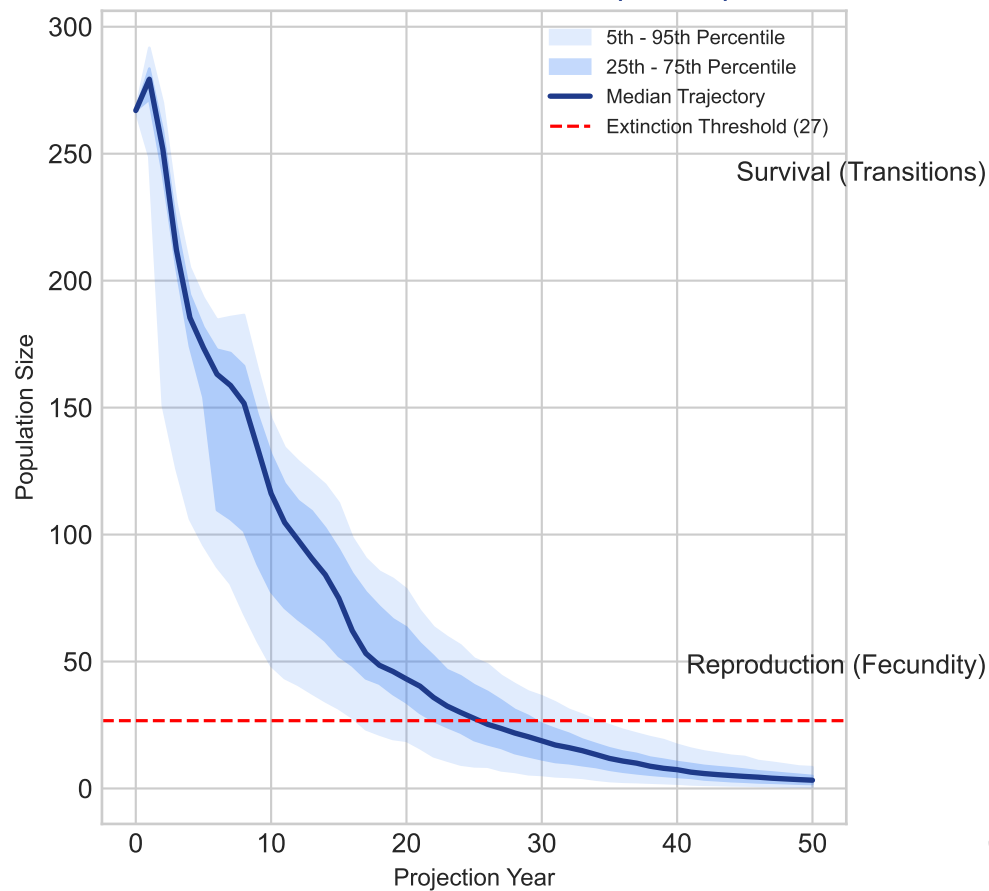


MSY Linear Program Solved (Yield: 0.0)



Environmental Stochasticity & Vital Rate Elasticities

Monte Carlo Extinction Risk (100.0%)



Elasticity Contribution to Growth (λ_1)

